

Jenkins User Management

Class no:5

Jenkins User management:

Managing users in Jenkins involves creating, configuring, and maintaining user accounts for individuals who interact with the Jenkins instance. This page provides a step-by-step guide on how to manage users effectively.

In Jenkins, **User Management** is the system and process used to control who can access your Jenkins server, what they can see, and what actions they can perform.

Think of it as the security guard and the manager of your CI/CD pipeline. Without proper user management, anyone could accidentally delete a pipeline, view sensitive credentials, or trigger production deployments.

What managing users means

In Jenkins, managing users refers to the process of creating, configuring, and maintaining user accounts. This includes:

- Creating new users.
- Configuring user details such as full name, email address, and password.
- Modifying user information.
- Deleting users.

Administrators with the appropriate permissions can perform these actions.

Effective user management in Jenkins relies on two core pillars: **Authentication** (who are you?) and **Authorization** (what are you allowed to do?).

1. Authentication (Who are you?)

Authentication is the process of verifying a user's identity. Jenkins provides a few ways to handle this:

- **Jenkins' Own User Database:** Jenkins maintains its own internal list of users and passwords. This is great for small teams or local testing.
- **LDAP / Active Directory:** For larger organizations, Jenkins can integrate with existing corporate directories. Users log in using their standard company credentials.
- **Single Sign-On (SSO) / OAuth:** You can use plugins to allow users to log in via GitHub, GitLab, Google, or Okta.

2. Authorization (What can you do?)

Once Jenkins knows *who* you are, authorization determines your access level. You configure this under **Manage Jenkins** -> **Security**.

There are three main authorization strategies used in Jenkins:

A. Matrix-based Security

This creates a large grid (matrix) where you manually check boxes to grant specific permissions (like Read, Discover, Configure, Administer) to individual users or groups.

- *Best for:* Small to medium teams who need precise control but don't have too many users to manage manually.

B. Role-Based Strategy (Via Plugin)

This is the most popular and scalable method for production environments. It requires the **Role-based Authorization Strategy** plugin. It breaks permissions down into three levels:

- **Global Roles:** Overall permissions (e.g., Admin, Read-only).
- **Item/Project Roles:** Permissions specific to certain jobs or pipelines based on naming patterns (e.g., a "QA Role" that can only run jobs starting with test-).
- **Agent Roles:** Permissions to manage specific build machines (nodes).

C. Folder-based Authorization

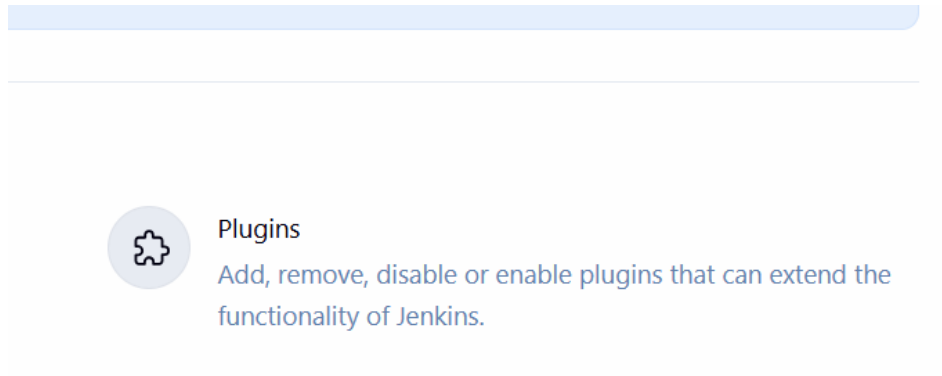
If you organize your Jenkins jobs into folders, you can assign permissions at the folder level. For example, the "Dev Team" can have full access to the "Development" folder, but zero access to the "Production" folder.

Summary of Key Permissions

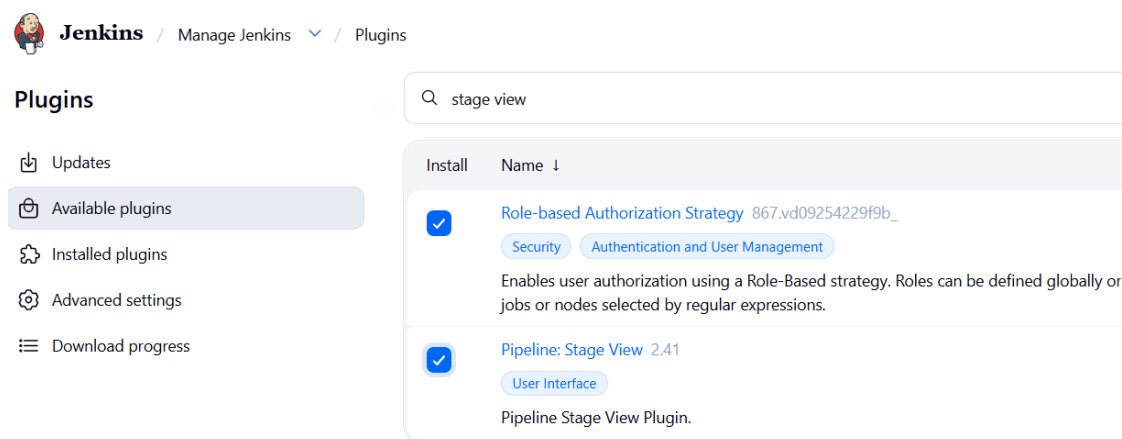
Permission Category	What it Allows
Administer	Full control over Jenkins (installing plugins, changing system settings).
Read	Allows the user to log in and view the Jenkins dashboard and job names.
Build	Allows the user to trigger/run a pipeline or job.
Configure	Allows the user to modify the settings, scripts, or steps of a specific job.
Workspace	Allows the user to look at the files pulled down during a build.

Hands on: Creating users and adding respective access over the Jenkins server

Step no:1 Login to Jenkins → **manage Jenkins** → choose **plugins**.



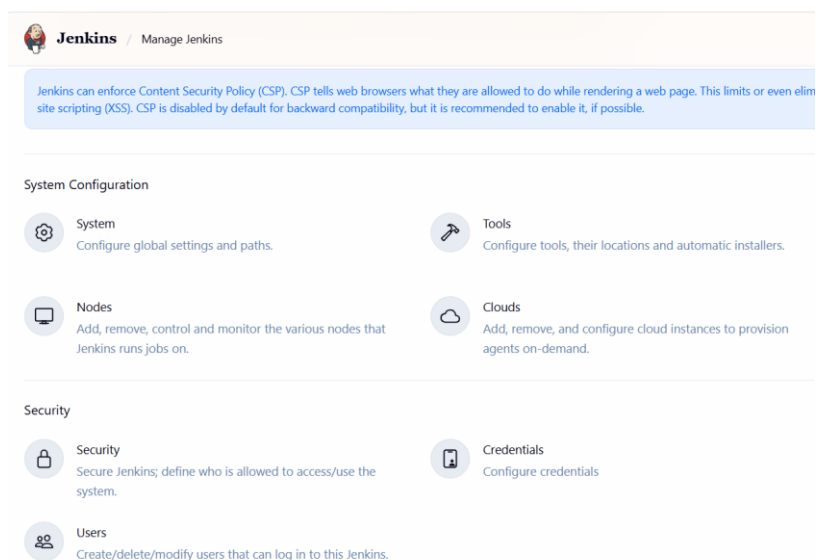
Available plugins → choose: **Role-based authorization strategy & Stage view.**



Step no:2 Log in to your Jenkins instance with an account that has administrative permissions.

Navigate to the **Manage Jenkins** page from the Jenkins dashboard.

Select **Users** under the **Security** section.



Step no:3 To create a new user in Jenkins:

On the **Users** page, select **Create User**.

enkins / Manage Jenkins ▾ / Jenkins' own user database



Users 1

+ Create User

These users can log into Jenkins. This is a sub set of [this list](#), which also contains auto-created users who really just made some commits on some projects and have no direct Jenkins access.

User ID ↓	Name
 nithya	Nithya Sri 

Fill in the required details:

- **Username:** A unique identifier for the user.
- **Password:** A secure password for the user.
- **Confirm Password:** Re-enter the password.
- **Full Name:** The full name of the user.
- **Email Address:** The email address of the user.

I am creating three user developer, test and support

Create User

Username

developer

Password

.....

Confirm password

.....

Full name

developer

E-mail address

dev@gmail.com

Create User

Select **Create User** to save the new user.

User ID ↓	Name		
 developer	developer		
 nithya	Nithya Sri		
 support	support		
 test	test		

Step no:4 Go to manage Jenkins → security →authorization: role-based strategy

Authorization

Role-Based Strategy

Markup Formatter

Markup Formatter ?

Plain text

Save

Apply

Step no:5 Now go to the manage and assign roles.

Security



Security

Secure Jenkins; define who is allowed to access/use the system.



Credentials

Configure credentials



Manage and Assign Roles

Handle permissions by creating roles and assigning them to users/groups



Users

Create/delete/reset users

Add our respective user we have created and provide the roles (access) for the users.

Manage and Assign Roles / Manage Roles



Role		Overall	Credentials				Agent					Job					Run		View		SCM	Metrics											
		Administer	Manage	Read	Create	Delete	Manage Domains	Update	View	Build	Configure	Connect	Create	Delete	Disconnect	Provision	Build	Cancel	Configure	Create	Delete	Discover	Read	Workspace	Delete	Replay	Update	Configure	Create	Delete	Read	Tag	HealthCheck
 admin																																	
 developer																																	
 test																																	
 support																																	

Role to add

Step no:6 To check the permissions, login as developer (user), see the console.

We can see the permissions being denied.

Sign in to Jenkins

Username

developer

Password

.....

☐ Keep me signed in

Sign in

EVS CHAP 1.pdf Dell View Ads | VuexyB... Gmail YouTube Maps New Tab file systems in os -



Jenkins

Access Denied

! developer is missing the Overall/Read permission